Project: Worked examples
Title: Conductor sizing and overload coordination
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1. Scope and method

Calculates the design current, the smallest preferred device rating and the smallest conductor whose tabulated capacity carries that rating. It checks I_b ≤ I_n ≤ I_z and I_2 ≤ 1.45 I_z, using your own capacity and rating records.

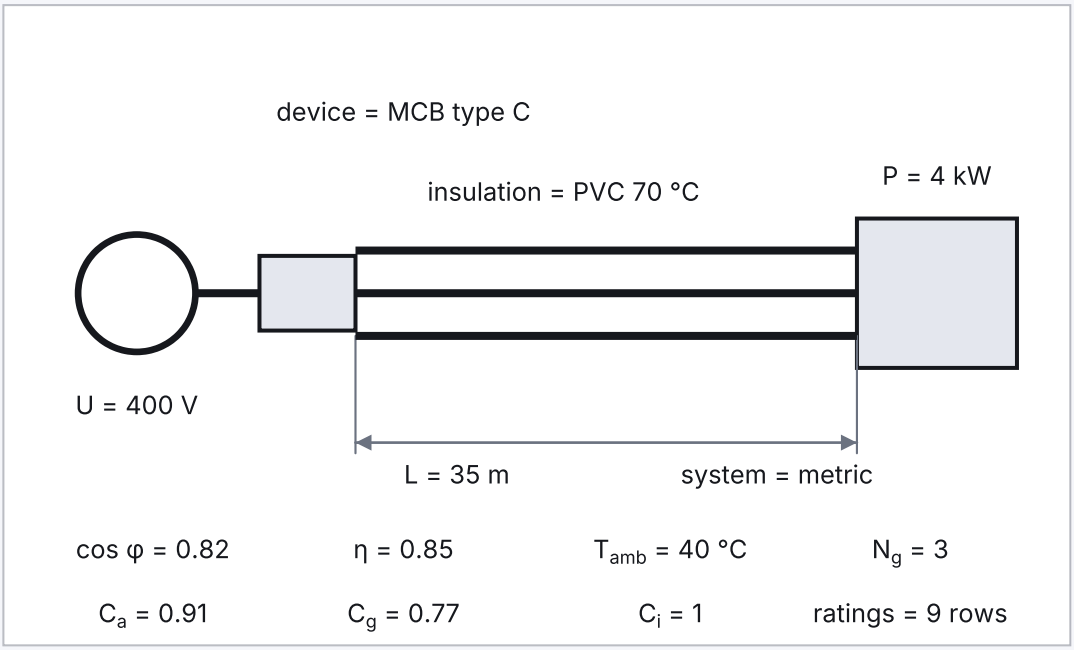


Figure 1 — A three-phase circuit: the supply U, the protective device, three line conductors of length L, and the load P. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Design current	I_b = P/(√3·U·cosφ·η) three-phase, P/(U·cosφ·η) single-phase, P/U DC; or the linked I_b
Protective device rating	I_n = smallest preferred rating in the ratings record with I_n ≥ I_b
Device and installation factors	C_f = 0.725 for a semi-enclosed fuse (1.45/2, derived), else 1; C_c = the entered factor for a cable buried direct or in ducts, 1 when none applies
Required tabulated capacity	$I_{t,req}$ = I_n/(C_a·C_g·C_i·C_f·C_c); optional derived C_a = √[(θ_p − θ_a)/(θ_p − θ_{ref})] (derived approximation — compare with the licensed table)
Conductor cross-section	CSA = smallest cross-section in the ampacity record with I_t ≥ $I_{t,req}$; AWG gauge n: d = 0.127·92^((36 − n)/39) mm, area = π d²/4 (derived)
Carrying capacity	I_z = I_t·C_a·C_g·C_i

WHAT IT CALCULATES	RELATION
Device and conductor coordination	$I_b \le I_n \le I_z$ and $I_2 \le 1.45 \cdot I_z$, $I_2 = 1.45 \cdot I_n$ inherent for MCB types B, C, D; $I_2 = 2 \cdot I_n$ by default for a semi-enclosed fuse; an entered I_2 (datasheet) is always checked; in the BS 7671 installation context a gG fuse to BS 88 (and an MCCB or MPCB entered without I_2, taken as $I_2 \le 1.45 \cdot I_n$) has the I_2 condition deemed satisfied when $I_n \le I_z$; in the machine-wiring context (IEC 60269-1 gG fuses have $I_2 = 1.6 \cdot I_n$, not $1.45 \cdot I_n$) I_2 must be entered
Device code	1 = MCB type B, 2 = type C, 3 = type D, 4 = fuse, MCCB or MPCB (carried to fault-protection so the fault check uses the device chosen here)

Basis: Co-ordination of conductors and overload protective devices[1](reg. 433.1.1); Current-carrying capacity of a cable[1](reg. 523.1); Current-carrying capacity of conductors and cables in normal service[2](cl. 12.4); Overcurrent protection[2](cl. 7.2); Circuit-breakers for overcurrent protection: conventional tripping current[3](cl. (number to be checked)); Overload protection by gG fuses and semi-enclosed fuses: deemed conditions[1](reg. 433.1.201, 433.1.202); Rating factors for the protective device type and for a cable buried in the ground or in ducts[1](App. 4 §5 (section number to be checked against the owner's copy)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
context	Design context (changes clause citations and the scope note only)	BS EN 60204-1 machine wiring		spec: Machine design basis MDB-2: BS EN 60204-1 wiring	any
supply	Supply	three-phase		drawing: E-SCH-021 rev A	any
device	Protective device	MCB type C		drawing: E-SCH-021 rev A	any
insulation	Insulation	PVC 70 °C		spec: Cable schedule CS-21: PVC 70 °C single cores	any
system	Conductor system (AWG: geometry only, no AWG capacities are supplied)	metric		spec: Cable schedule CS-21: metric cross-sections	any
method	Installation method key (descriptive only, e.g. B2, C, E)	B2		drawing: E-LAY-021 rev B	any
ratings	Preferred device ratings	9 row(s)		reference: record: owner's licensed table (not shipped); ratings are the typical preferred series	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
ampacity	Current-carrying capacities for the installation method	5 row(s)		reference: record: owner's licensed table (not shipped); capacities are synthetic values invented for this example	any
L	Circuit length (carried to the voltage-drop calculation)	35	m	drawing: E-LAY-021 rev B	1 … 300 m
P	Rated output (shaft) power; η converts it to input (DC: electrical power drawn, η not applied)	4	kW	spec: Motor schedule MS-04: conveyor drive 4 kW rated output	0.1 … 200 kW
U	Supply voltage (line voltage for three-phase)	400	V	spec: Site supply: 400 V three-phase	12 … 690 V
$\cos_{phi}$	Power factor (not used for DC)	0.82		spec: Motor datasheet MD-04: 0.82 at full load	0.6 … 1
η	Efficiency (not used for DC)	0.85		spec: Motor datasheet MD-04: 85 % at full load	0.6 … 1
T_{amb}	Ambient temperature	40	°C	spec: Machine design basis MDB-2: 40 °C inside the panel and trunking	0 … 60 °C
T_{ref}	Reference ambient temperature for the derived C_a	30	°C	assumption: Reference ambient of the licensed table; only used when C_a is derived.	20 … 40 °C
N_g	Grouped circuits	3		drawing: E-LAY-021 rev B	1 … 10
C_a	Ambient correction factor (1 does not apply the ambient: enter the factor for T_{amb} from your licensed table)	0.91		reference: record: owner's licensed table (not shipped); 0.91 is a synthetic value invented for this example	0.5 … 1.3
C_g	Grouping correction factor (at most 1)	0.77		reference: record: owner's licensed table (not shipped); 0.77 is a synthetic value invented for this example	0.3 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
C_i	Thermal insulation correction factor (at most 1)	1		reference: record: owner's licensed table (not shipped); 1 is a neutral value for this example	0.3 … 1
C_c	Buried or ducted installation factor, at most 1 (1 does not apply it: enter the factor for a cable buried direct or in ducts from your licensed table)	1		default	0.5 … 1
$derive_{Ca}$	Derive C_a from the temperatures instead of the entered factor	no		assumption: The entered C_a is used; the derived approximation is not applied.	any

TABLE ROW	VALUES
ratings[1]	I_n = 6 A
ratings[2]	I_n = 10 A
ratings[3]	I_n = 16 A
ratings[4]	I_n = 20 A
ratings[5]	I_n = 25 A
ratings[6]	I_n = 32 A
ratings[7]	I_n = 40 A
ratings[8]	I_n = 50 A
ratings[9]	I_n = 63 A
ampacity[1]	CSA = 1.5 mm²; I_t = 12.3 A
ampacity[2]	CSA = 2.5 mm²; I_t = 17.7 A
ampacity[3]	CSA = 4 mm²; I_t = 24.1 A
ampacity[4]	CSA = 6 mm²; I_t = 31.9 A
ampacity[5]	CSA = 10 mm²; I_t = 44.6 A

3. Assumptions

A1. Every tabulated value is entered from the user's own licensed copy or a manufacturer datasheet. (No table from a standard or a manufacturer is shipped.)
A2. No table from a standard or a manufacturer ships with the calculator. (No table from a standard or a manufacturer is shipped.)
A3. The capacities in the table are for the installation method stated. (Records carry their source and edition.)
A4. The method key is descriptive and is not interpreted. (Records carry their source and edition.)
A5. A gG fuse is deemed to meet $I_2 \le 1.45 \cdot I_z$ when $I_n \le I_z$ only under BS 7671. (BS 7671 reg. 433.1.201; IEC 60269-1.)

A6. An IEC 60269-1 gG fuse has $I_2 = 1.6 \cdot I_n$, so in the machine-wiring context I_2 is entered from the datasheet. (BS 7671 reg. 433.1.201; IEC 60269-1.)
A7. The ambient, grouping, insulation, fuse and installation factors are applied as a product. (The count is carried to the sheet and warns when $C_g = 1$ with more than one circuit.)
A8. C_g is not looked up from the number of grouped circuits. (The count is carried to the sheet and warns when $C_g = 1$ with more than one circuit.)
A9. The optional derived C_a is an approximation for the temperature difference only. (Derived, not tabulated.)
A10. The derived C_a is labelled as an approximation, so compare it with the licensed table. (Derived, not tabulated.)
A11. $C_f = 0.725$ applies to a semi-enclosed fuse only: 1.45 divided by an assumed fusing factor of 2. (Derived.)
A12. C_c is the factor for a cable buried direct or in ducts, entered from the user's licensed table. (BS 7671 App. 4 names the factor; its value is licensed and is not shipped.)
A13. AWG areas follow the geometry d = 0.127·92^((36 – n)/39) mm. (Derived.)
A14. No AWG capacity is supplied. (Derived.)
A15. DC circuits use $I_b = P/U$. (Method.)
A16. The power factor and the efficiency are not applied to a DC circuit. (Method.)
A17. T_{ref} (Reference ambient temperature for the derived C_a) is taken as 30 °C. (Reference ambient of the licensed table; only used when C_a is derived.)
A18. $derive_{Ca}$ (Derive C_a from the temperatures instead of the entered factor) is taken as no. (The entered C_a is used; the derived approximation is not applied.)
A19. Machine wiring context (BS EN 60204-1): the conductor is sized for the protective device rating; the capacities and factors are the user's records for the installation method stated. (The context changes citations, and whether a gG fuse, MCCB or MPCB without I_2 may be deemed to meet $I_2 \le 1.45 \cdot I_z$ (installation) or must have I_2 entered (machine wiring); no other number differs.)
A20. No factor for a cable buried direct or in ducts is applied ($C_c = 1$). (Enter C_c from your licensed table for a cable buried direct or in ducts.)

4. Working

[14] $I_b = P / (\sqrt{3} * U * cos_{phi} * \eta)$
$= 4\ kW / (\sqrt{3} \times 400\ V \times 0.82 \times 0.85)$
$= 8.2834\ A$
Design current (three-phase)

[15] I_n = smallest preferred rating with $I_n \ge I_b$ ²(cl. 7.2)
= smallest rating ≥ 8.2834 A in the ratings record
= 10 A
Rated current of the protective device (ratings record)

[16] $device_{code}$ = 1 = MCB type B, 2 = type C, 3 = type D, 4 = fuse, MCCB or MPCB ²(cl. 7.2)
= MCB type C
= 2
Protective device type code (carried to E5 so the fault check uses this device)

[17] C_f = 1 (not a semi-enclosed fuse) ²(cl. 12.4)
= MCB type C
= 1
Semi-enclosed fuse factor

[22] $T_{op} = 70\ degC$
$= 70\ degC$
Conductor operating temperature for the insulation

[25] $I_{t,req} = I_n / (C_a * C_g * C_i * C_f * C_c)$ ²(cl. 12.4)
$= 10\ A / (0.91 \times 0.77 \times 1 \times 1 \times 1)$
$= 14.271\ A$

Required tabulated capacity

[26] I_t = I_t of the selected row of the ampacity record [2](cl. 12.4)

= $I_{t,req}$ = 14.271 A

= 17.7 A

Tabulated current-carrying capacity of the selected conductor

[27] CSA = smallest cross-section in the ampacity record with $I_t \geq I_{t,req}$ [2](cl. 12.4)

= $I_{t,req}$ = 14.271 A

= 2.5 mm²

Selected conductor cross-section

[28] $I_z = I_t * C_a * C_g * C_i$ [2](cl. 12.4)

= 17.7 A × 0.91 × 0.77 × 1

= 12.402 A

Current-carrying capacity in the installed conditions

[29] $u_{Ib} = I_b / I_n$ [2](cl. 7.2; BS 7671 reg. 433.1.1)

= 8.2834 A / 10 A

= 0.82834

Design current utilisation of the device

[30] $u_{In} = I_n / I_z$ [2](cl. 7.2; BS 7671 reg. 433.1.1)

= 10 A / 12.402 A

= 0.8063

Device rating utilisation of the conductor

[31] $I2 = 1.45 * I_n$ [2](cl. 7.2; BS 7671 reg. 433.1.1)

= 1.45 × 10 A

= 14.5 A

Conventional operating current (inherent 1.45 I_n)

[32] $I2_{lim} = 1.45 * I_z$ [2](cl. 7.2; BS 7671 reg. 433.1.1)

= 1.45 × 12.402 A

= 17.983 A

Largest conventional operating current the conductor allows, 1.45 I_z

[33] $u_{I2} = I2 / I2_{lim}$ [2](cl. 7.2; BS 7671 reg. 433.1.1)

= 14.5 A / 17.983 A

= 0.8063

I_2 utilisation of 1.45 I_z

5. Results

SYMBOL	VALUE	DESCRIPTION
I_b	8.283 A	Design current (governing)
I_n	10 A	Rated current of the device
CSA	2.500 mm²	Selected cross-section
I_z	12.40 A	Current-carrying capacity
L	35 m	Circuit length
T_{op}	70 °C	Conductor operating temperature for the insulation
$device_{code}$	2	Protective device type code
C_f	1	Semi-enclosed fuse factor
C_c	1	Buried or ducted installation factor
C_a	0.91	Ambient factor
$I_{t,req}$	14.27 A	Required tabulated capacity
I2	14.50 A	Conventional operating current

SYMBOL	VALUE	DESCRIPTION
u_{Ib}	0.83	I_b / I_n
u_{In}	0.81	I_n / I_z
u_{I2}	0.81	$I_2 / (1.45\ I_z)$

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Design current against the device rating	8.283 A	10 A	0.83	²(cl. 7.2; BS 7671 reg. 433.1.1)
PASS	Device rating against the conductor capacity	10 A	12.40 A	0.81	²(cl. 7.2; BS 7671 reg. 433.1.1)
PASS	Conventional operating current against 1.45 times the conductor capacity	14.50 A	17.98 A	0.81	²(cl. 7.2; BS 7671 reg. 433.1.1)

7. Applicability

WARNING records: tabulated values not taken from a reference record are ampacity (Cable ampacity (current-carrying capacities)); C_a, C_g, C_i (Cable correction factors); ratings (Preferred device ratings). Enter each as one so its source, edition and licence travel with it. [record]

Defaults used (inputs left empty):
C_c: not entered, so the default of 1 is used.

8. Verification

```
Template elec.conductor-sizing@1.5.0 Content hash c0657fdab51c927d Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. British Standards Institution; Institution of Engineering and Technology. Requirements for Electrical Installations: IET Wiring Regulations. BS 7671. IET.
2. BS EN 60204-1:2018.
3. BS EN 60898-1:2019.